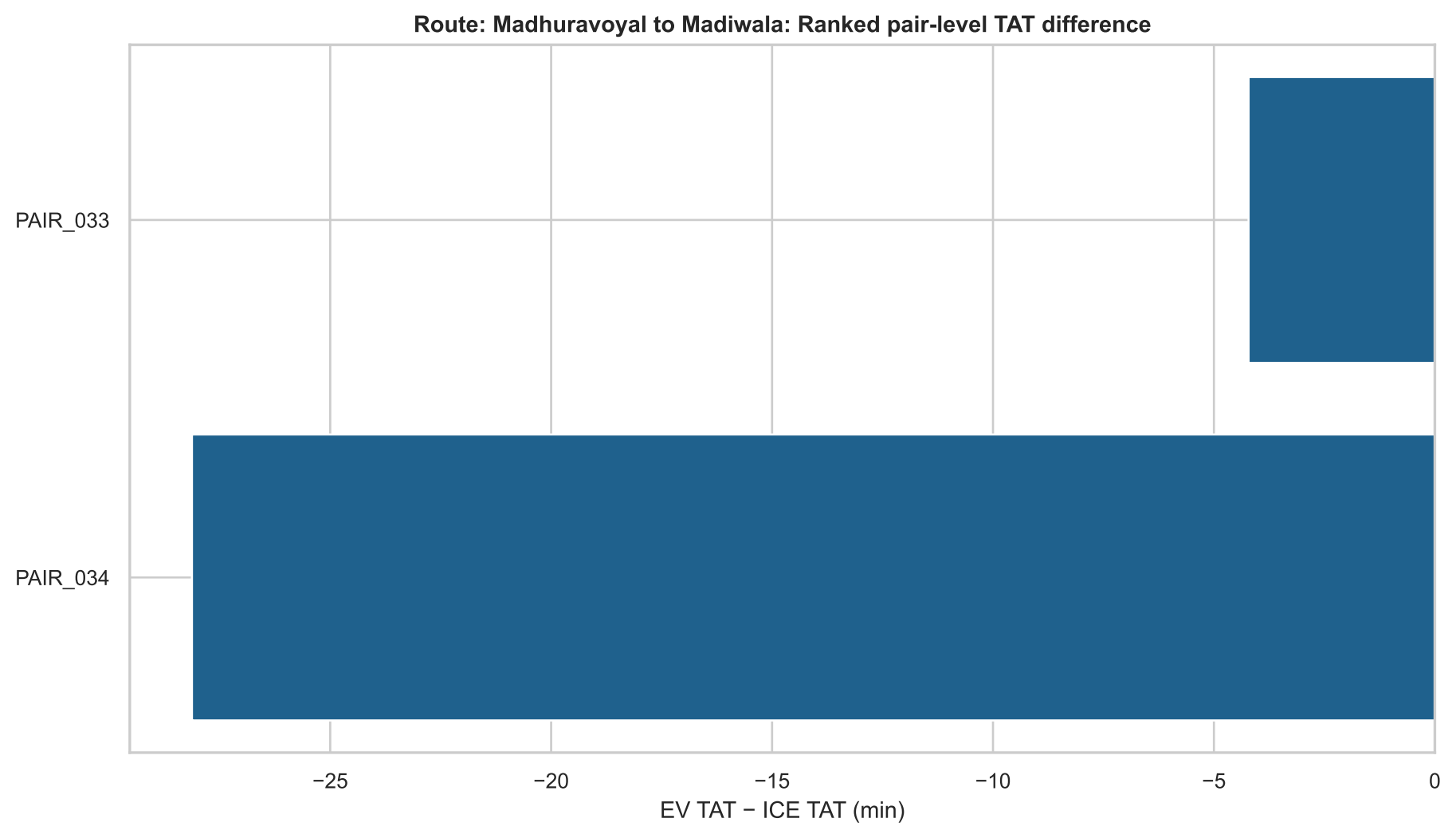
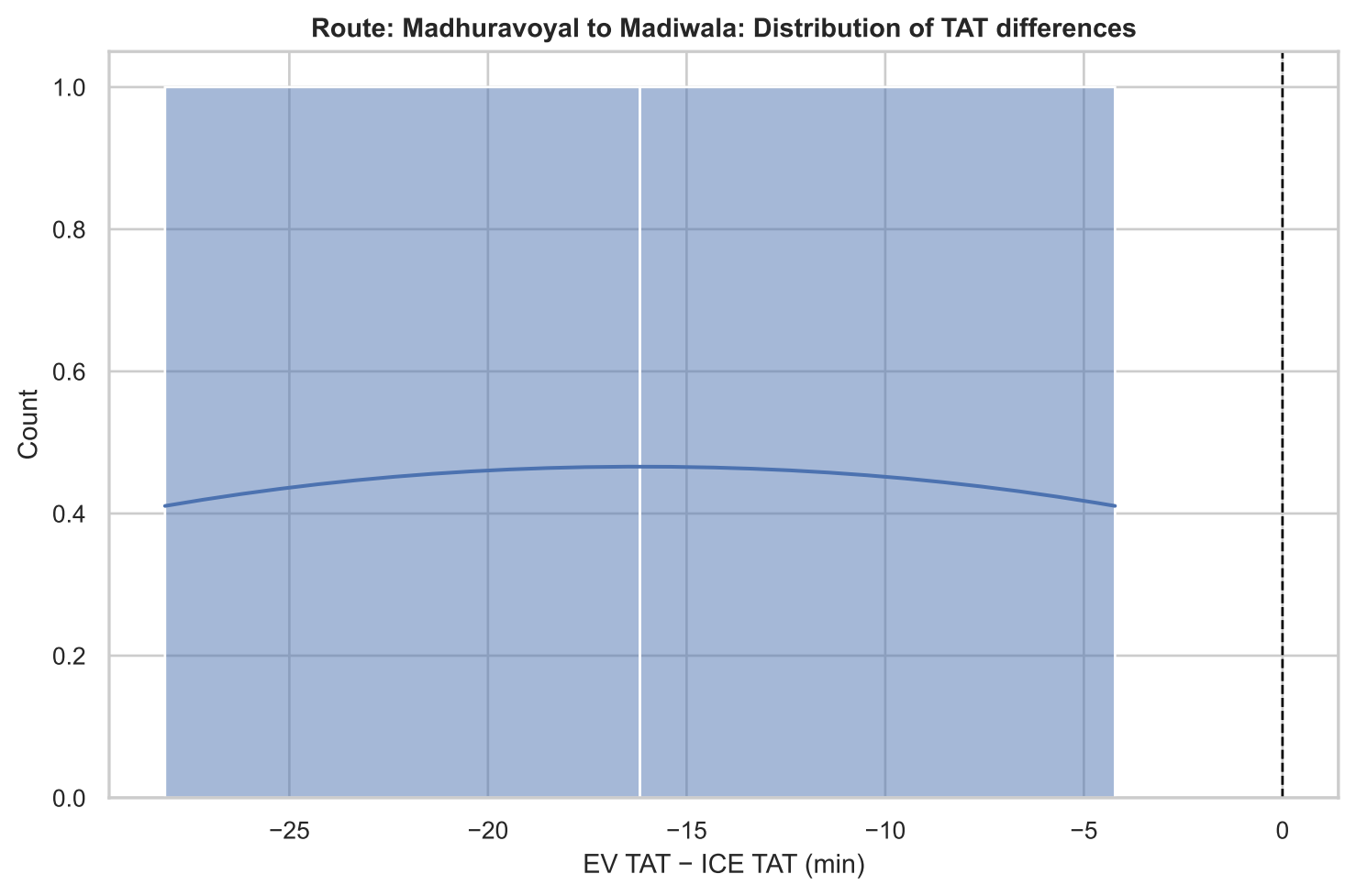
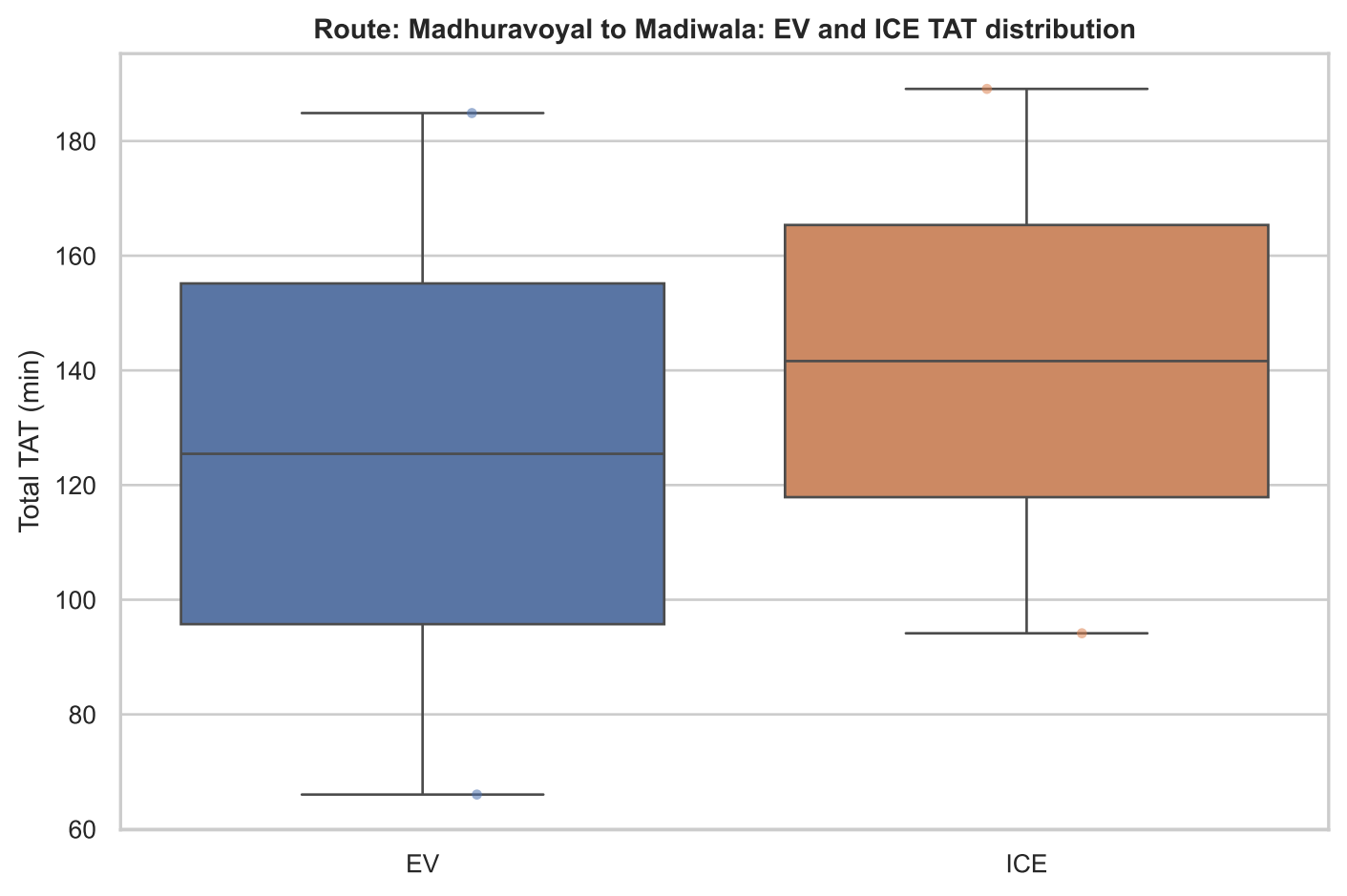
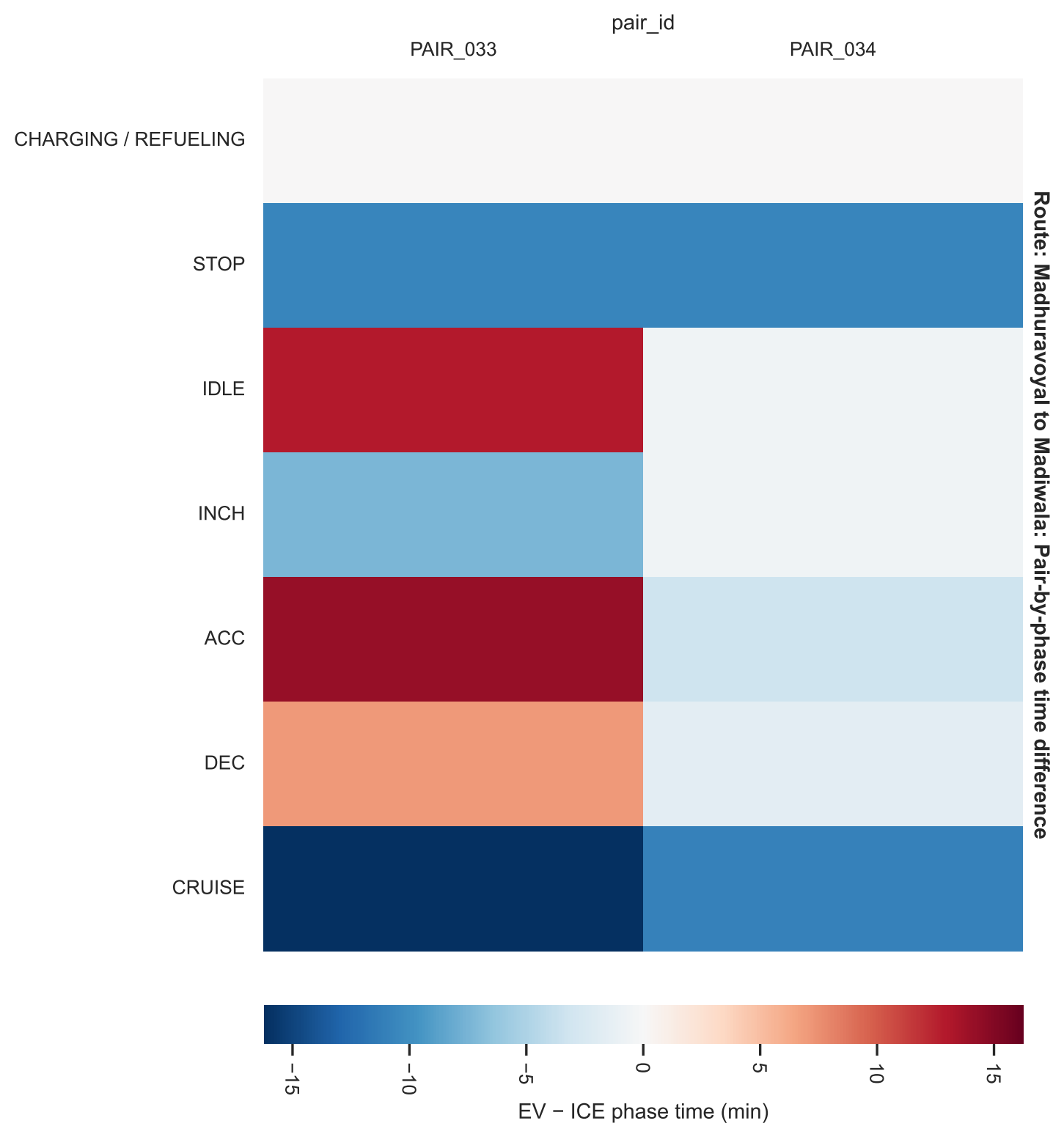
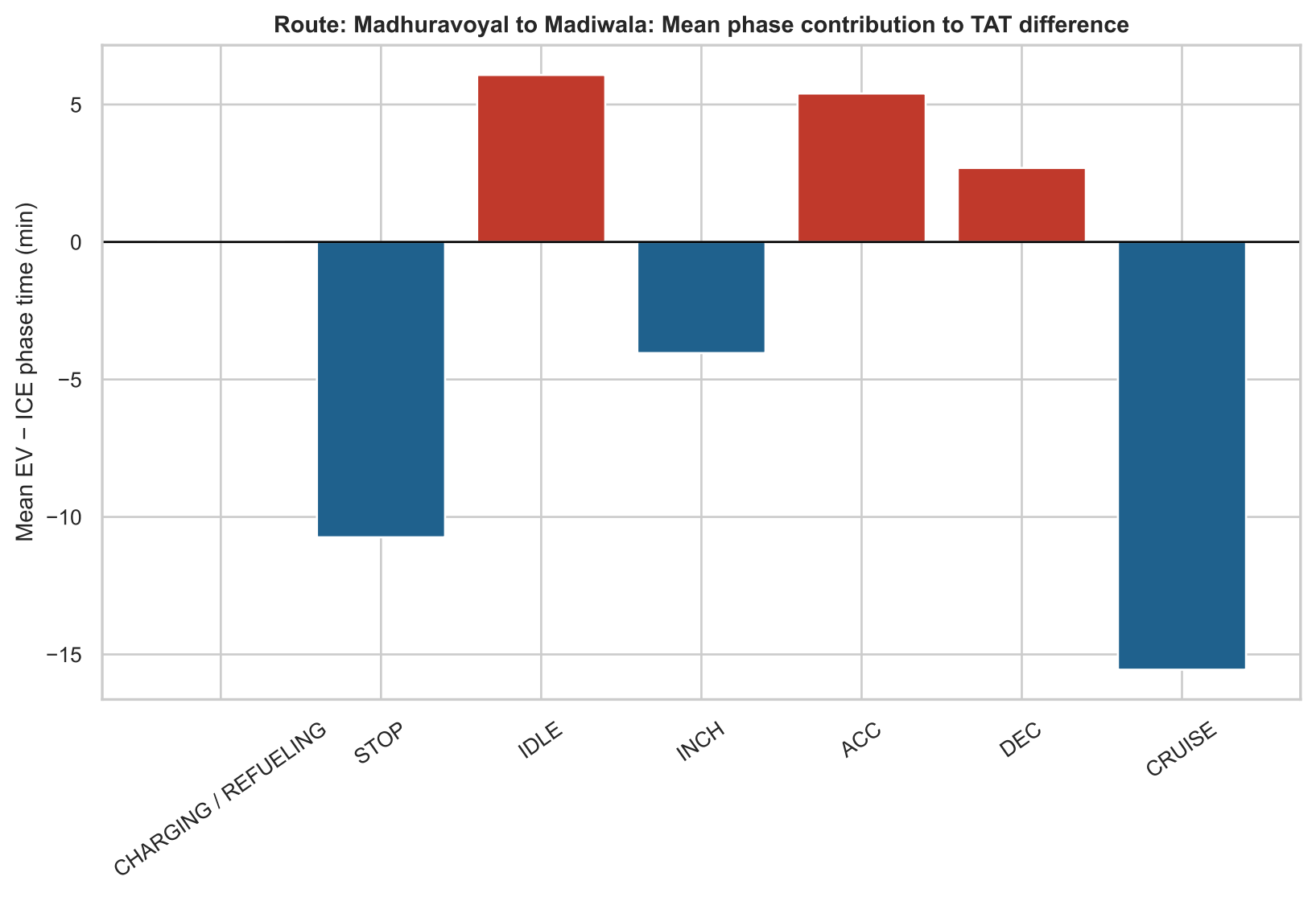
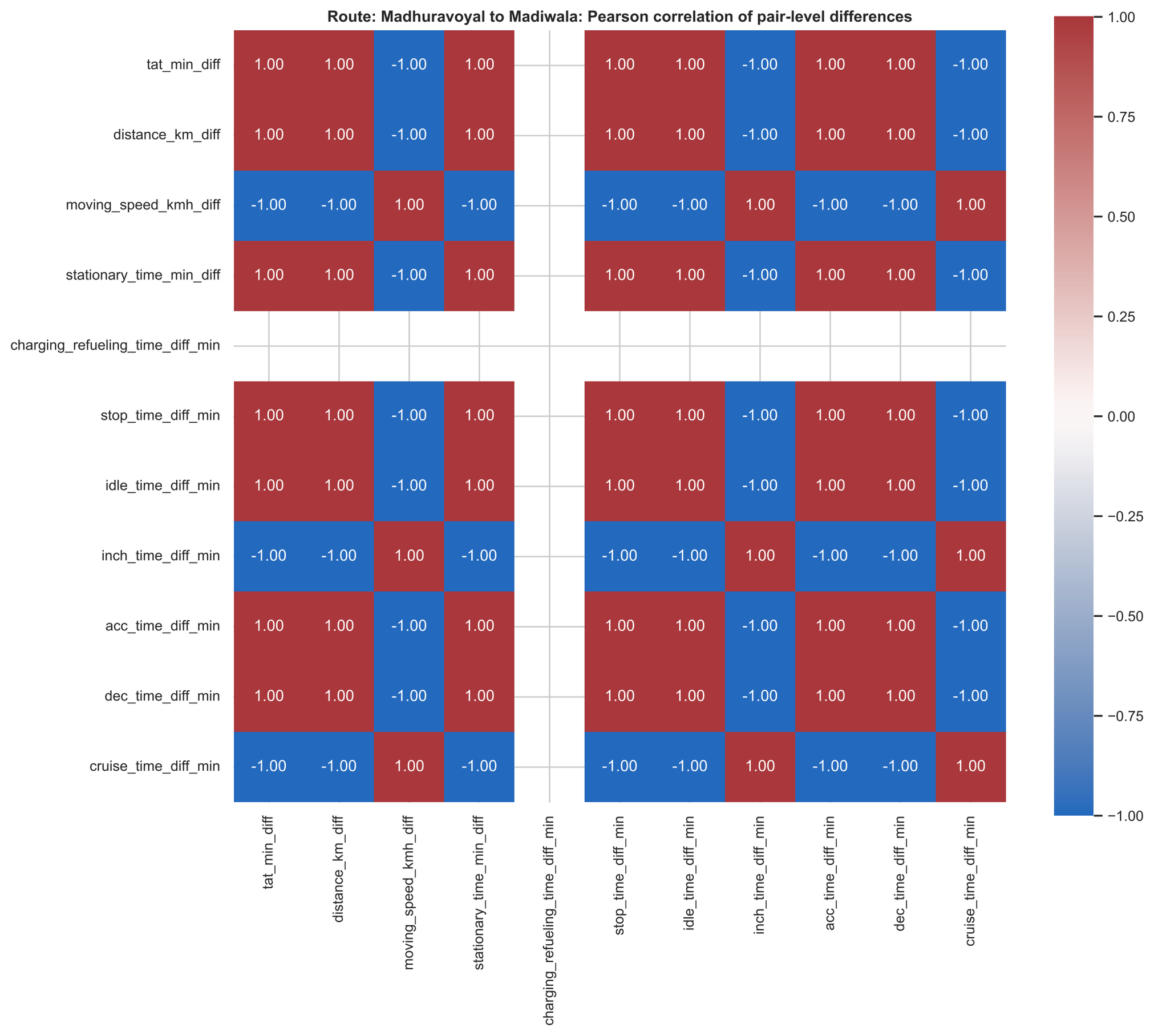

Route: Madhuravoyal to Madiwala: Ranked pair-level TAT difference
PAIR_033
PAIR_034
−25
−20
−15
−10
−5
0
EV TAT − ICE TAT (min)

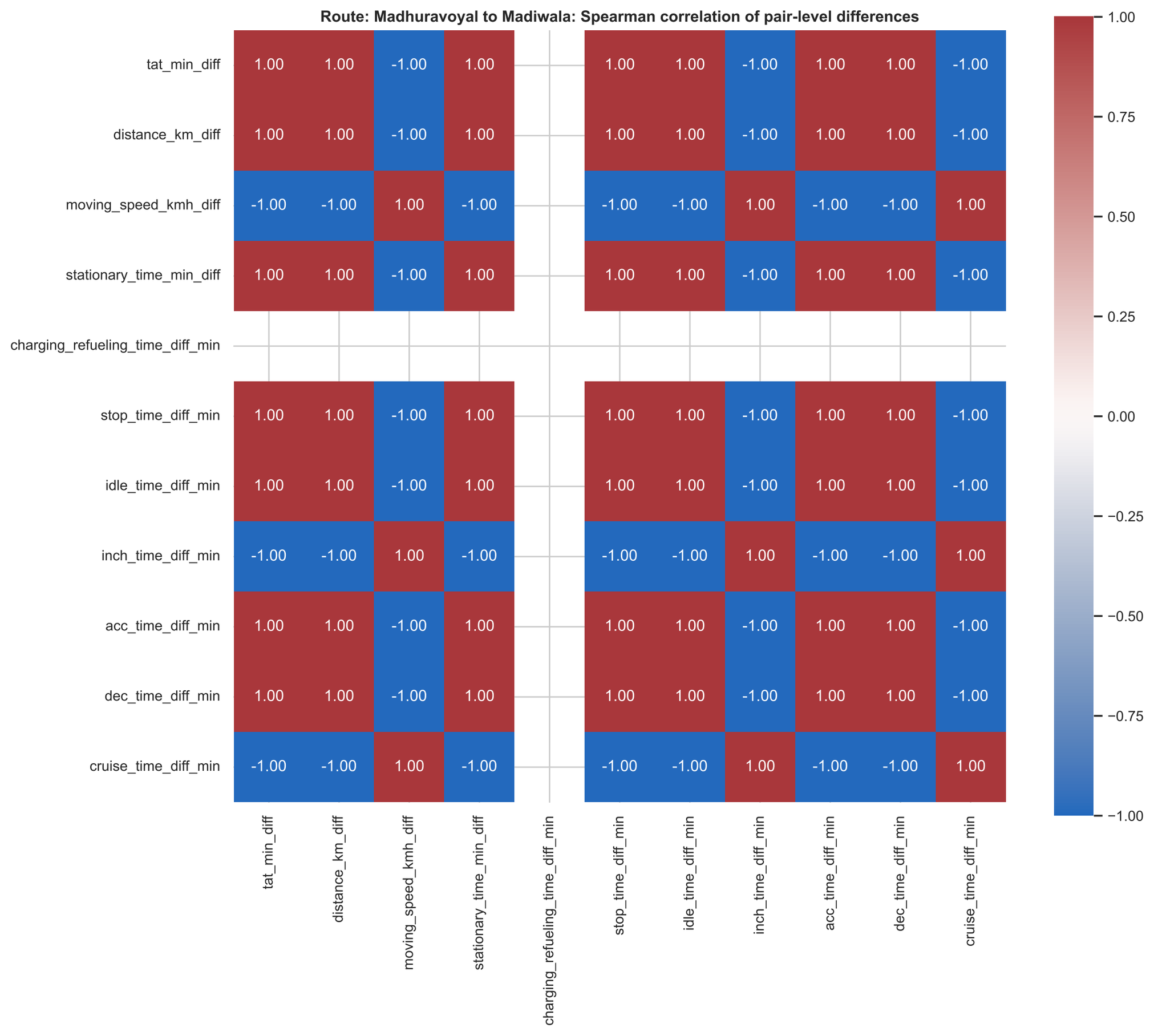

Route: Madhuravoyal to Madiwala: Distribution of TAT differences
Count
1.0
0.8
0.6
0.4
0.2
0.0
−25
−20
−15
−10
−5
0
EV TAT − ICE TAT (min)

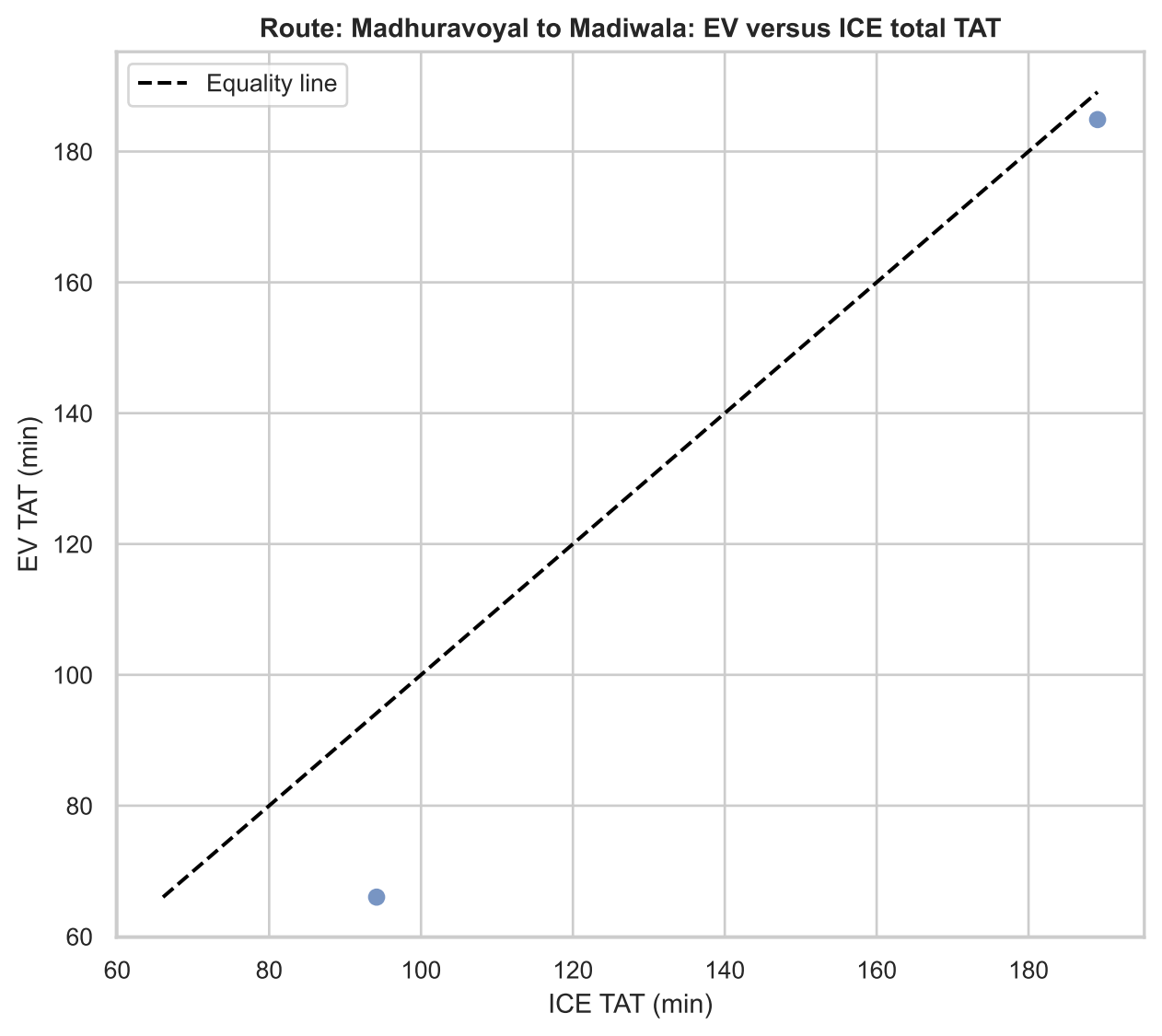

Route: Madhuravoyal to Madiwala: EV and ICE TAT distribution
Total TAT (min)
180
160
140
120
100
80
60
EV
ICE

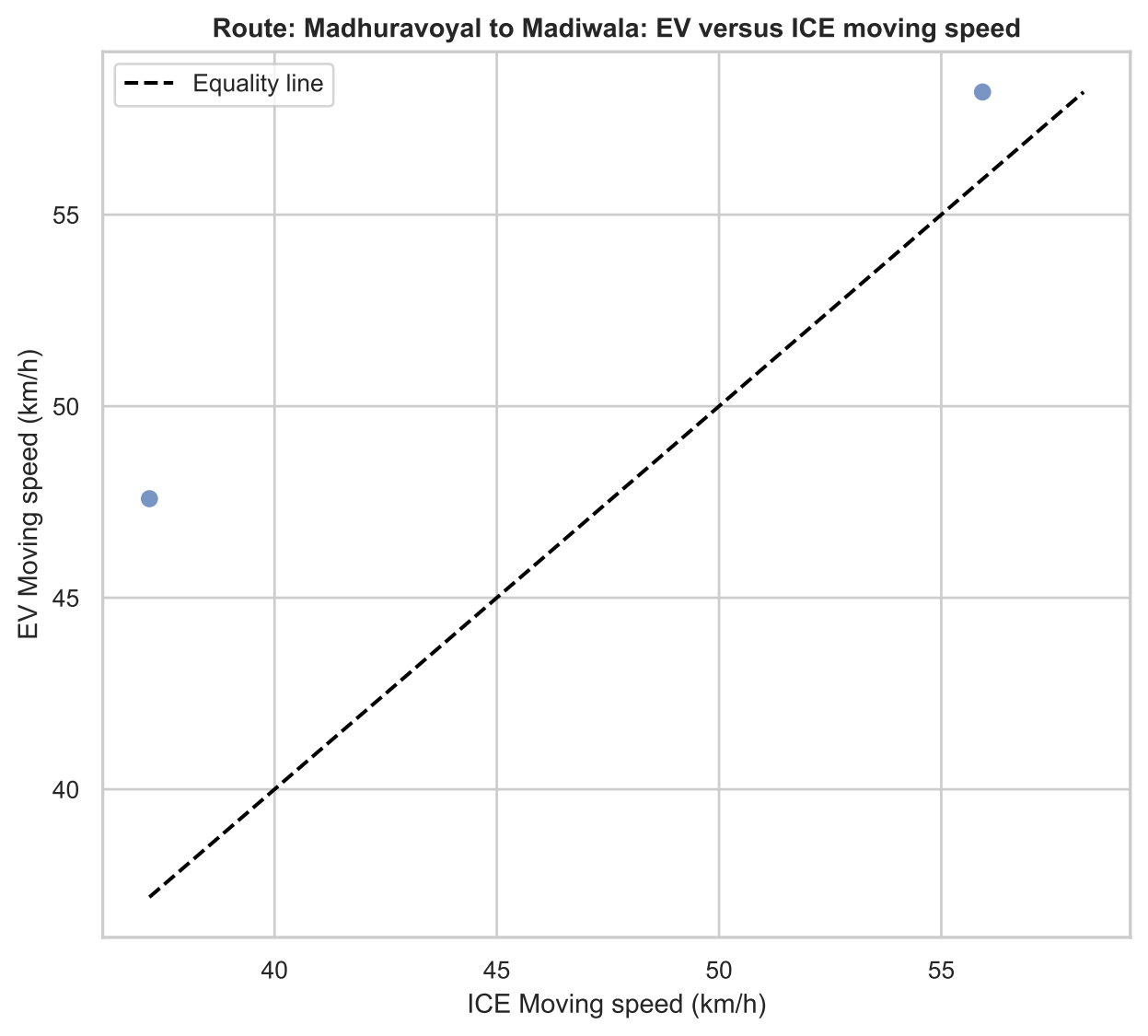

pair_id
PAIR_033
PAIR_034
CHARGING / REFUELING
STOP
IDLE
INCH
ACC
DEC
CRUISE
Route: Madhuravoyal to Madiwala: Pair-by-phase time difference
−15
−10
−5
0
5
10
15
EV − ICE phase time (min)

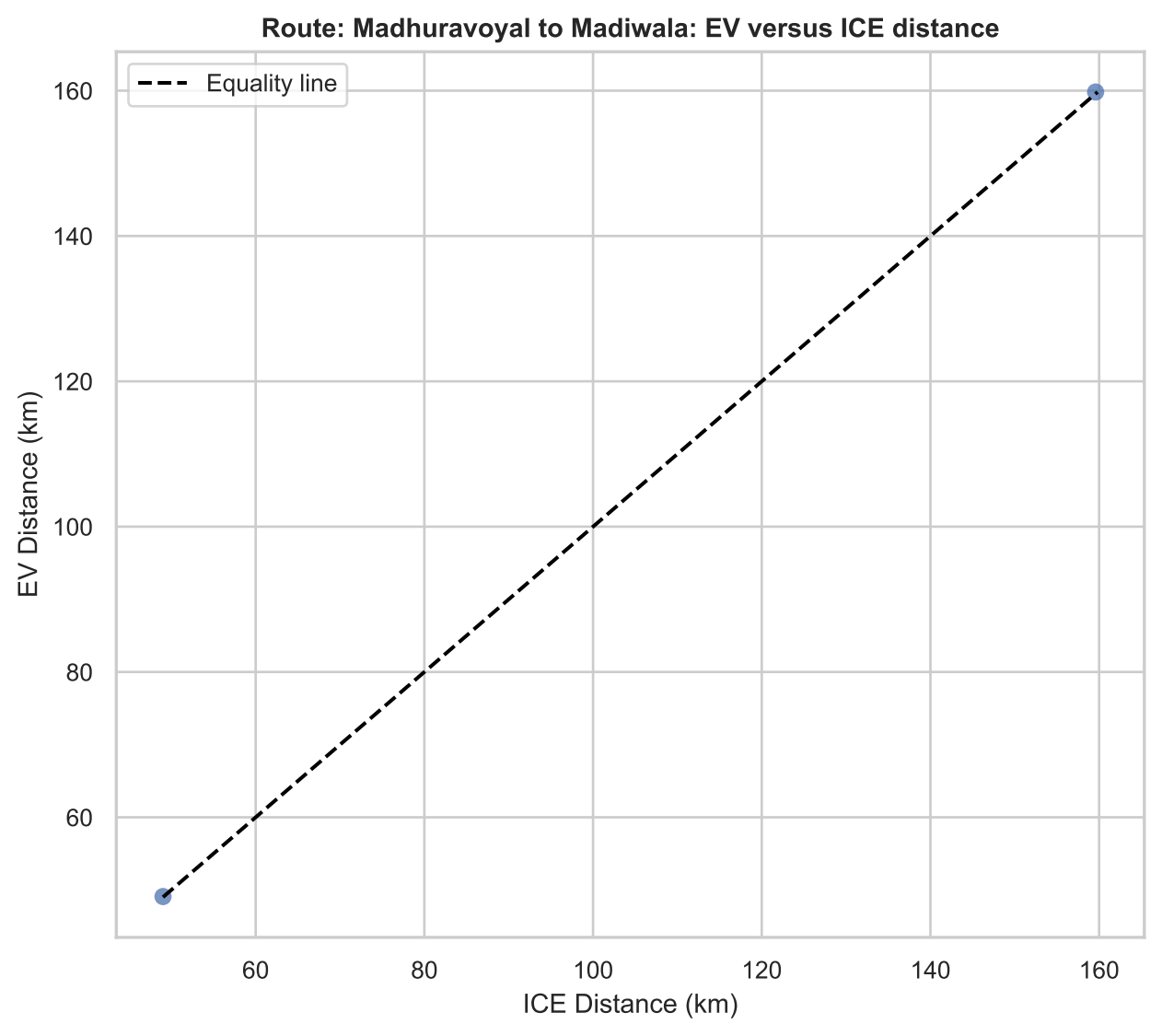

Route: Madhuravoyal to Madiwala: Mean phase contribution to TAT difference
Mean EV – ICE phase time (min)
5
0
−5
−10
−15
CHARGING / REFUELING
STOP
IDLE
INCH
ACC
DEC
CRUISE

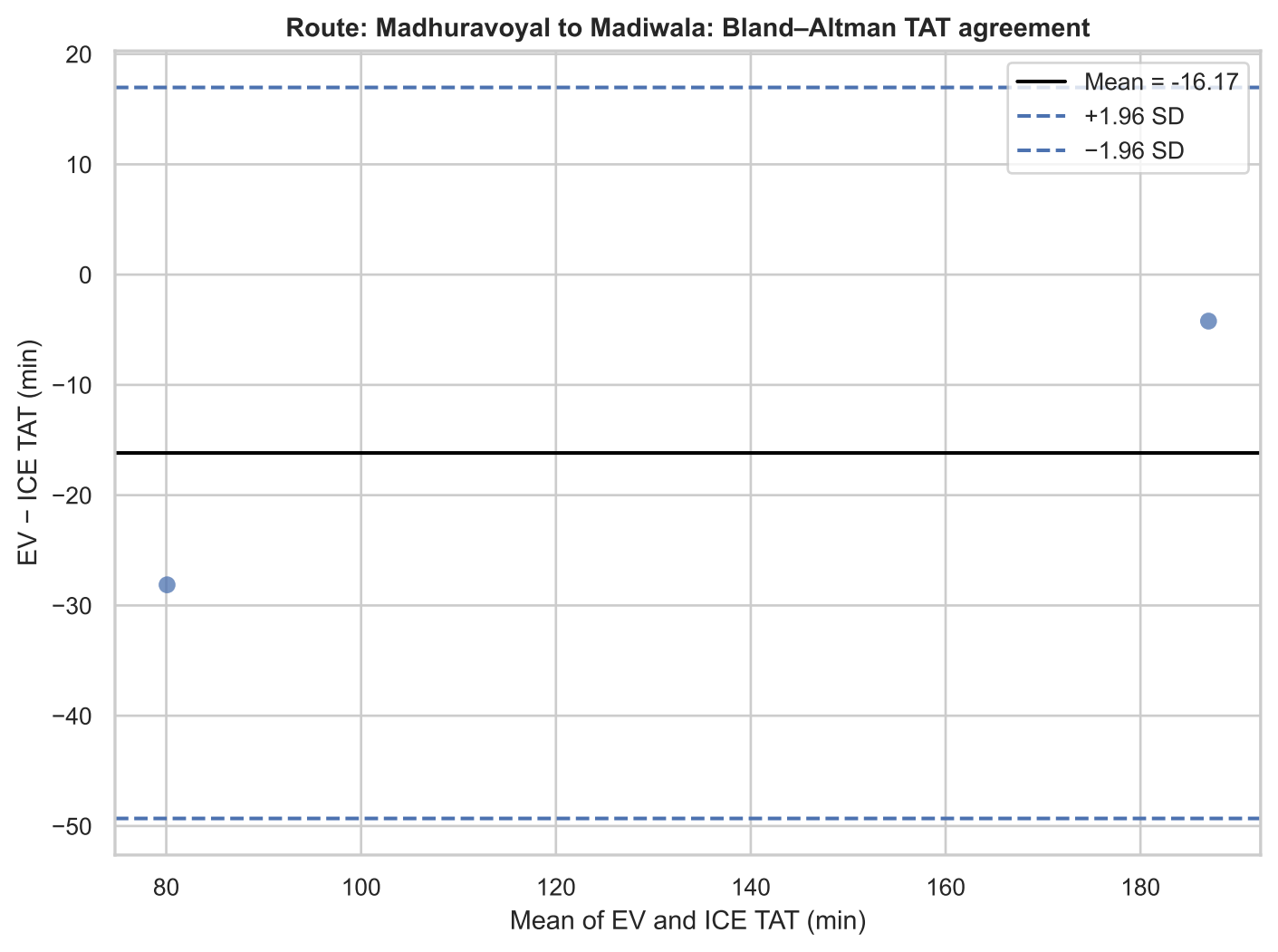

Route: Madhuravoyal to Madiwala: Pearson correlation of pair-level differences
tat_min_diff
distance_km_diff
moving_speed_kmh_diff
stationary_time_min_diff
charging_refueling_time_diff_min
stop_time_diff_min
idle_time_diff_min
inch_time_diff_min
acc_time_diff_min
dec_time_diff_min
cruise_time_diff_min
1.00
0.75
0.50
0.25
0.00
−0.25
−0.50
−0.75
−1.00

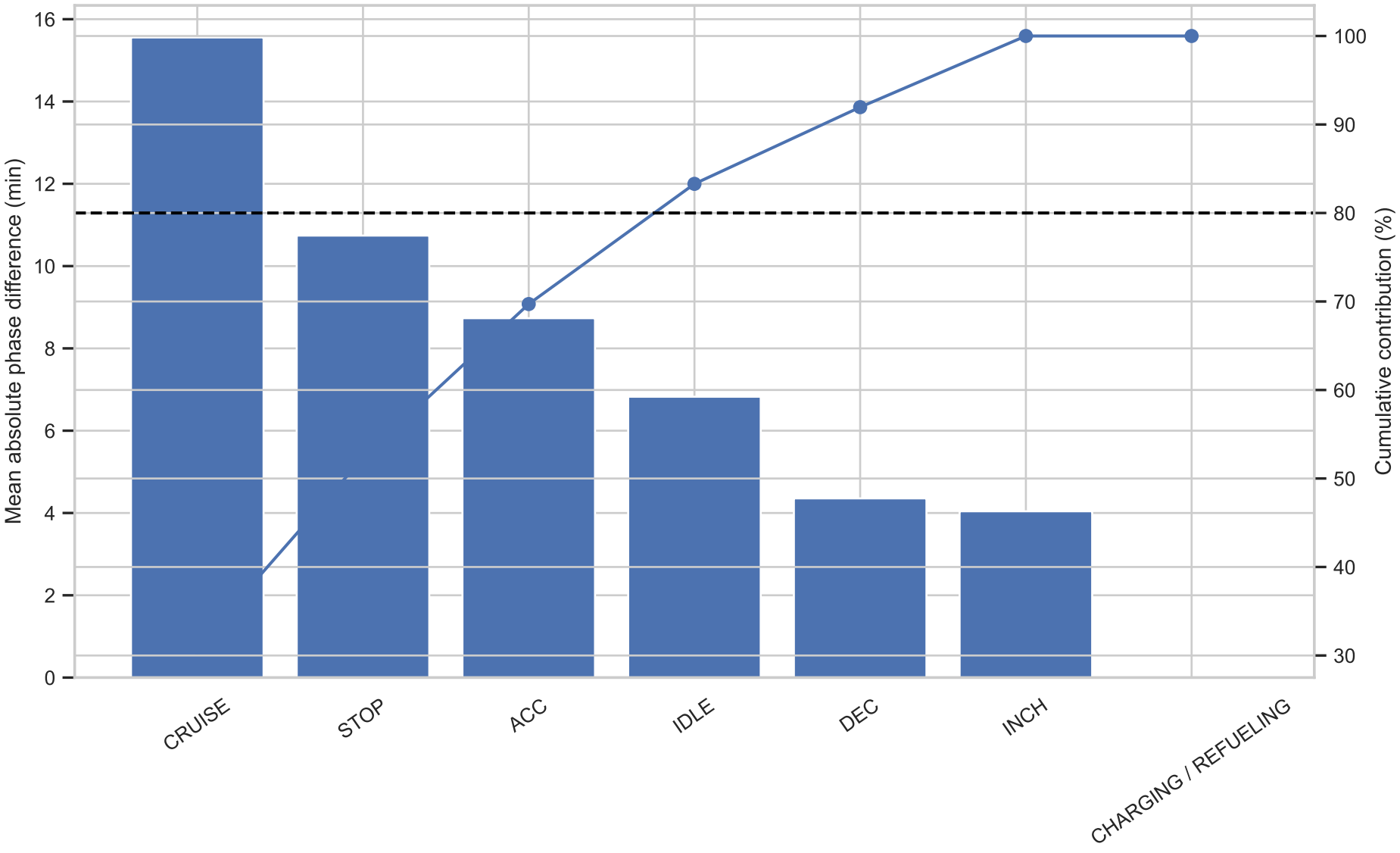

Route: Madhuravoyal to Madiwala: Spearman correlation of pair-level differences
tat_min_diff
distance_km_diff
moving_speed_kmh_diff
stationary_time_min_diff
charging_refueling_time_diff_min
stop_time_diff_min
idle_time_diff_min
inch_time_diff_min
acc_time_diff_min
dec_time_diff_min
cruise_time_diff_min
1.00
0.75
0.50
0.25
0.00
−0.25
−0.50
−0.75
−1.00

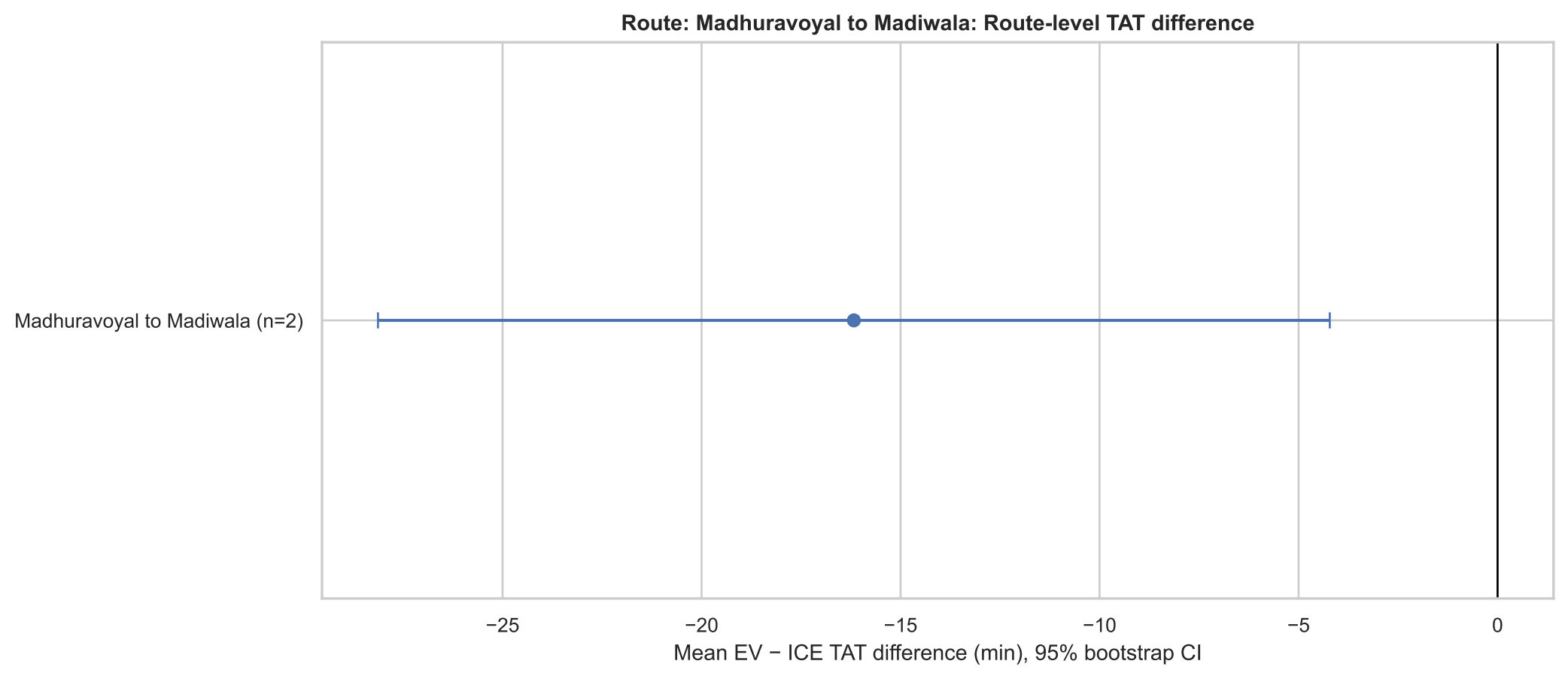

Route: Madhuravoyal to Madiwala: EV versus ICE total TAT
Equality line
EV TAT (min)
ICE TAT (min)
60
80
100
120
140
160
180

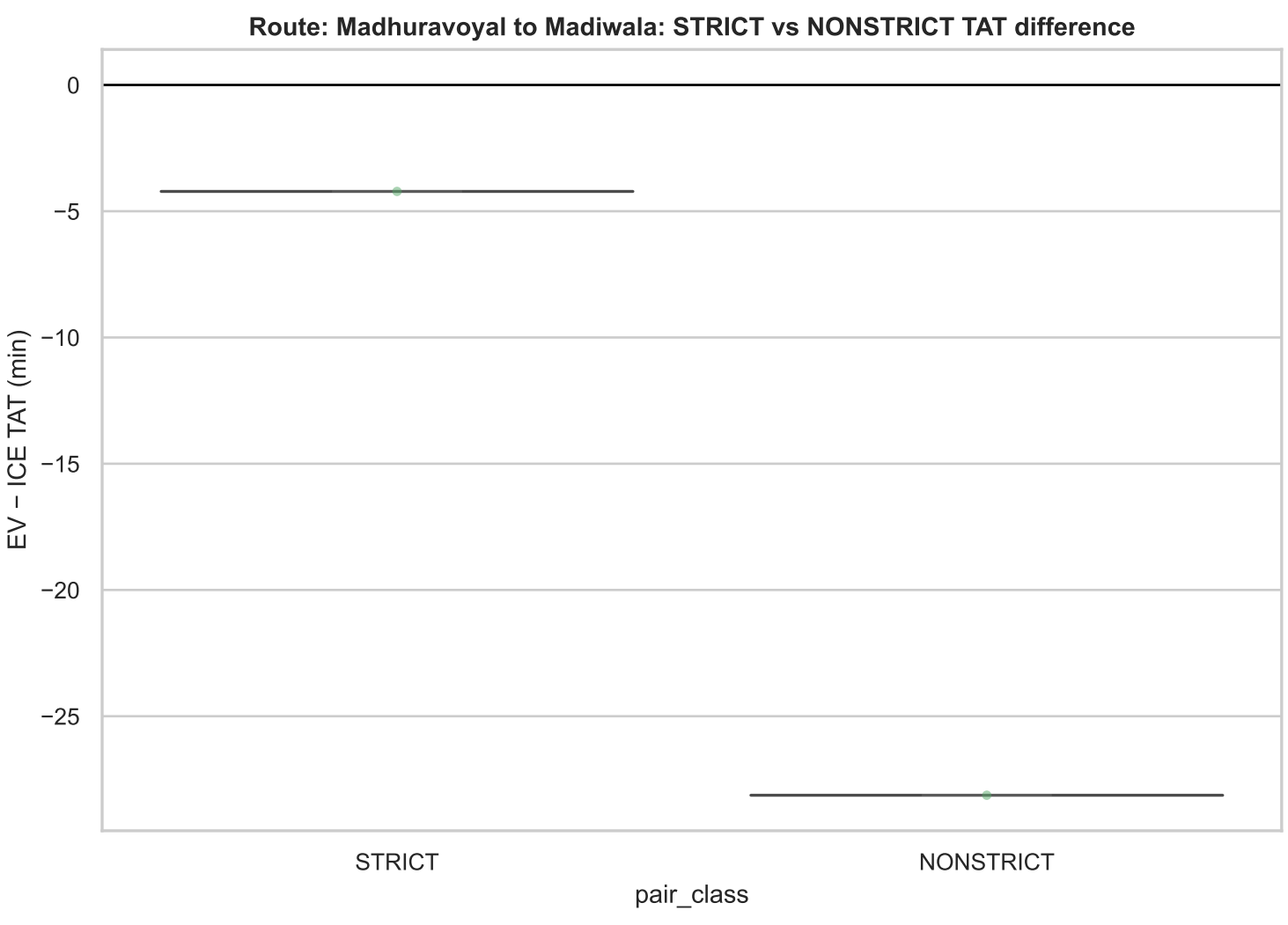

Route: Madhuravoyal to Madiwala: EV versus ICE moving speed
Equality line
EV Moving speed (km/h)
ICE Moving speed (km/h)
40
45
50
55
40
45
50
55

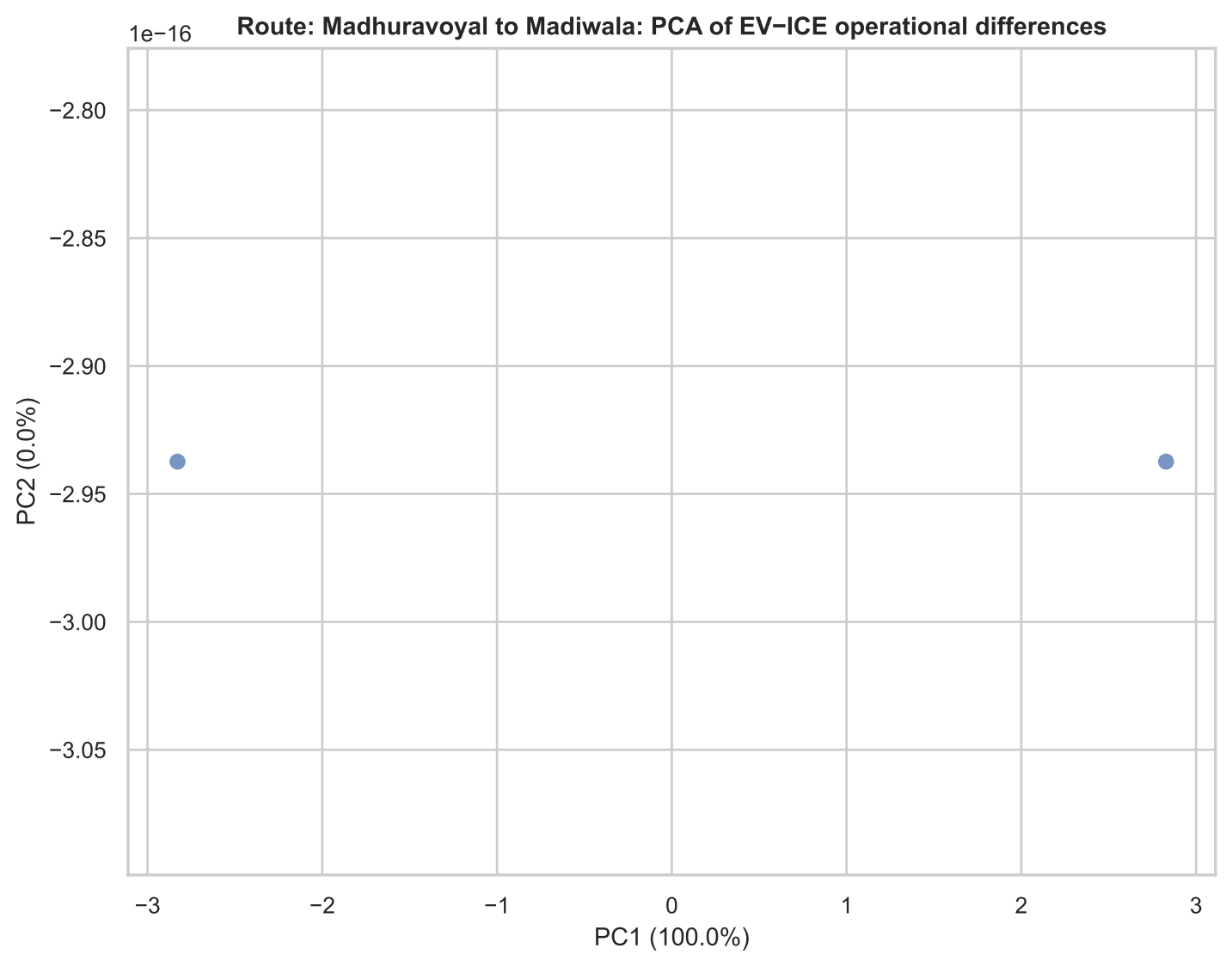

Route: Madhuravoyal to Madiwala: EV versus ICE distance
Equality line
EV Distance (km)
ICE Distance (km)
60
80
100
120
140
160

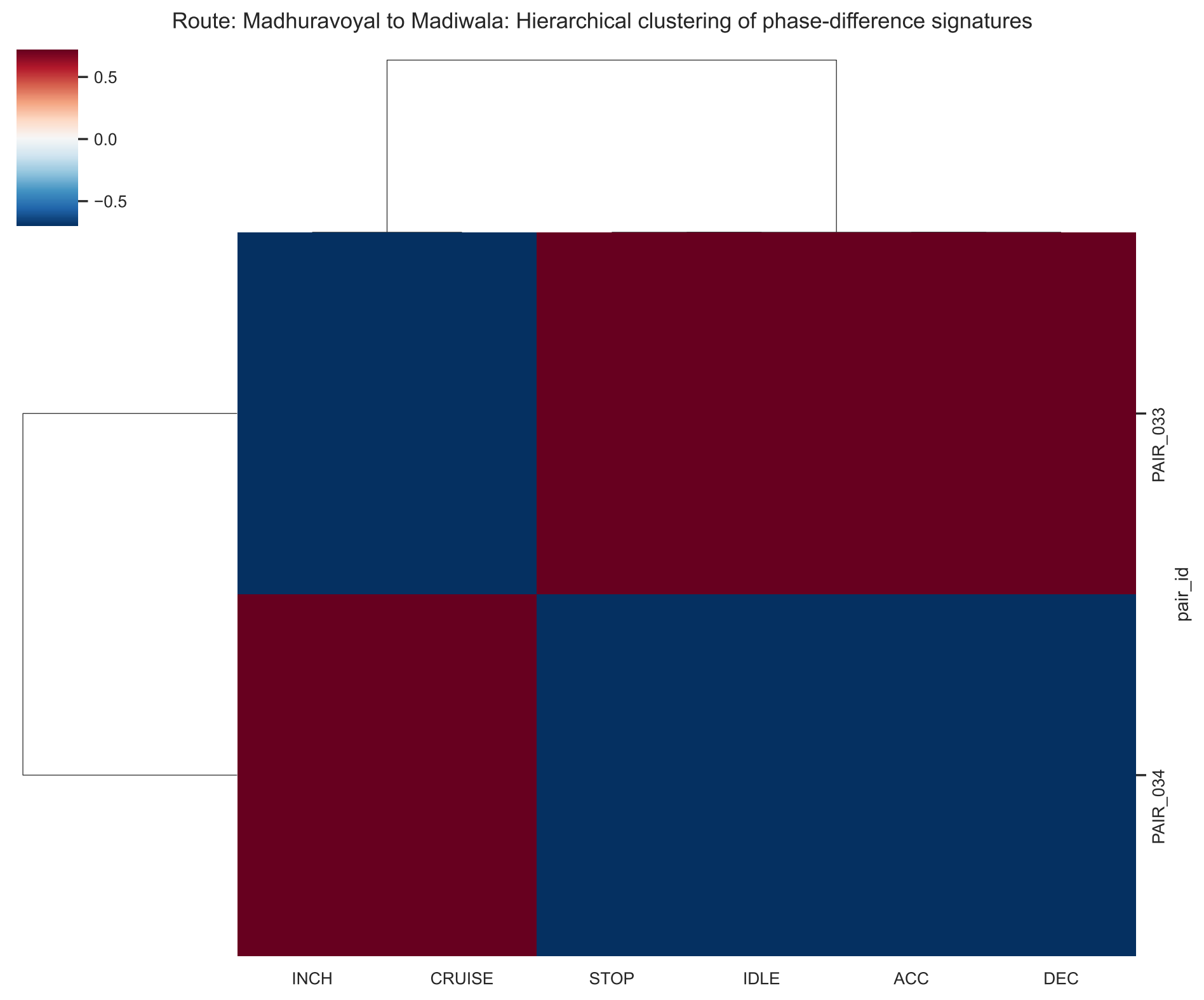

Route: Madhuravoyal to Madiwala: Bland–Altman TAT agreement
Mean = -16.17
+1.96 SD
−1.96 SD
20
10
0
−10
−20
−30
−40
−50
EV – ICE TAT (min)
80
100
120
140
160
180
Mean of EV and ICE TAT (min)


Route: Madhuravoyal to Madiwala: Pareto analysis of phase effects
Mean absolute phase difference (min)
Cumulative contribution (%)
0
2
4
6
8
10
12
14
16
30
40
50
60
70
80
90
100
CRUISE
STOP
ACC
IDLE
DEC
INCH
CHARGING / REFUELING


Route: Madhuravoyal to Madiwala: Route-level TAT difference
Madhuravoyal to Madiwala (n=2)
−25
−20
−15
−10
−5
0
Mean EV − ICE TAT difference (min), 95% bootstrap CI


Route: Madhuravoyal to Madiwala: STRICT vs NONSTRICT TAT difference
0
−5
−10
−15
−20
−25
EV − ICE TAT (min)
STRICT
NONSTRICT
pair_class


Route: Madhuravoyal to Madiwala: PCA of EV−ICE operational differences
1e−16
−2.80
−2.85
−2.90
−2.95
−3.00
−3.05
PC2 (0.0%)
−3
−2
−1
0
1
2
3
PC1 (100.0%)


Route: Madhuravoyal to Madiwala: Hierarchical clustering of phase-difference signatures
0.5
0.0
−0.5
PAIR_033
PAIR_034
pair_id
INCH
CRUISE
STOP
IDLE
ACC
DEC